# Excel Formulas & Rules — Complete Reference

## 1. Math & Trigonometry

| Formula | Purpose |
|---|---|
| `SUM(range)` | Adds values |
| `SUMIF(range, criteria, [sum_range])` | Sums with one condition |
| `SUMIFS(sum_range, crit_range1, crit1, ...)` | Sums with multiple conditions |
| `SUMPRODUCT(array1, array2, ...)` | Multiplies arrays, then sums |
| `PRODUCT(range)` | Multiplies values |
| `ABS(number)` | Absolute value |
| `ROUND(number, digits)` | Rounds to N digits |
| `ROUNDUP(number, digits)` / `ROUNDDOWN(...)` | Rounds up/down |
| `MROUND(number, multiple)` | Rounds to nearest multiple |
| `CEILING(number, sig)` / `FLOOR(number, sig)` | Rounds to nearest significance |
| `INT(number)` | Rounds down to integer |
| `TRUNC(number, [digits])` | Truncates decimal |
| `MOD(number, divisor)` | Remainder after division |
| `QUOTIENT(num, denom)` | Integer division result |
| `POWER(number, power)` | Exponentiation (same as `^`) |
| `SQRT(number)` | Square root |
| `EXP(number)` | e raised to power |
| `LN(number)` / `LOG(number, [base])` | Natural/other log |
| `GCD(...)` / `LCM(...)` | Greatest common divisor / least common multiple |

| Formula | Purpose |
| --- | --- |
| `RAND()` | Random decimal 0–1 |
| `RANDBETWEEN(bottom, top)` | Random integer in range |
| `PI()` | Value of π |
| `SIGN(number)` | Returns -1, 0, or 1 |
| `SUBTOTAL(function_num, range)` | Aggregates ignoring hidden/filtered rows |
| `AGGREGATE(function_num, options, ref)` | Aggregates with error/hidden-row handling |

## 2. Statistical

| Formula | Purpose |
|---|---|
| `AVERAGE(range)` | Mean |
| `AVERAGEIF(range, criteria, [avg_range])` | Mean with one condition |
| `AVERAGEIFS(avg_range, crit_range1, crit1, ...)` | Mean with multiple conditions |
| `MEDIAN(range)` | Middle value |
| `MODE.SNGL(range)` | Most frequent value |
| `MIN(range)` / `MAX(range)` | Smallest/largest value |
| `MINIFS(...)` / `MAXIFS(...)` | Min/Max with conditions |
| `COUNT(range)` | Counts numeric cells |
| `COUNTA(range)` | Counts non-empty cells |
| `COUNTBLANK(range)` | Counts empty cells |
| `COUNTIF(range, criteria)` | Counts with one condition |
| `COUNTIFS(range1, crit1, ...)` | Counts with multiple conditions |
| `STDEV.S(range)` / `STDEV.P(range)` | Standard deviation (sample/population) |
| `VAR.S(range)` / `VAR.P(range)` | Variance (sample/population) |
| `LARGE(range, k)` / `SMALL(range, k)` | Kth largest/smallest value |
| `RANK.EQ(number, range, [order])` | Rank within a range |
| `PERCENTILE.INC(range, k)` | Kth percentile |
| `QUARTILE.INC(range, quart)` | Quartile value |
| `CORREL(array1, array2)` | Correlation coefficient |
| `TREND(known_y, known_x, new_x)` | Linear trend forecast |
| `FORECAST.LINEAR(x, known_y, known_x)` | Predicts future value |

## 3. Text

| Formula | Purpose |
|---|---|
| `CONCAT(text1, text2, ...)` / `&` operator | Joins text |
| `TEXTJOIN(delim, ignore_empty, range)` | Joins with delimiter |
| `LEFT(text, n)` / `RIGHT(text, n)` | Extract chars from start/end |
| `MID(text, start, n)` | Extract chars from middle |
| `LEN(text)` | Character count |
| `TRIM(text)` | Removes extra spaces |
| `UPPER(text)` / `LOWER(text)` / `PROPER(text)` | Case conversion |
| `SUBSTITUTE(text, old, new, [instance])` | Replaces text |
| `REPLACE(text, start, n, new_text)` | Replaces by position |
| `FIND(find_text, text)` | Case-sensitive search position |
| `SEARCH(find_text, text)` | Case-insensitive search position |
| `TEXT(value, format)` | Formats a number as text |
| `VALUE(text)` | Converts text to number |
| `NUMBERVALUE(text)` | Converts text to number (locale-aware) |
| `EXACT(text1, text2)` | Case-sensitive comparison |
| `REPT(text, n)` | Repeats text n times |
| `CHAR(number)` / `CODE(text)` | Character ↔ code conversion |
| `TEXTSPLIT(text, delimiter)` | Splits text into cells (365) |
| `TEXTBEFORE(text, delim)` / `TEXTAFTER(text, delim)` | Extract text before/after delimiter (365) |

## 4. Logical

| Formula | Purpose |
|---|---|
| `IF(condition, value_if_true, value_if_false)` | Conditional logic |
| `IFS(cond1, val1, cond2, val2, ...)` | Multiple conditions, no nesting |
| `AND(cond1, cond2, ...)` | True if all conditions true |
| `OR(cond1, cond2, ...)` | True if any condition true |
| `NOT(condition)` | Reverses logical value |
| `IFERROR(value, value_if_error)` | Handles errors |
| `IFNA(value, value_if_na)` | Handles #N/A specifically |
| `SWITCH(expression, val1, result1, ...)` | Matches value to a result |
| `XOR(cond1, cond2, ...)` | True if odd number of conditions true |
| `TRUE()` / `FALSE()` | Logical constants |

## 5. Lookup & Reference

| Formula | Purpose |
|---|---|
| `VLOOKUP(value, table, col_index, [exact])` | Vertical lookup |
| `HLOOKUP(value, table, row_index, [exact])` | Horizontal lookup |
| `XLOOKUP(value, lookup_array, return_array, [if_na], [match_mode], [search_mode])` | Modern flexible lookup (365) |
| `LOOKUP(value, lookup_vector, result_vector)` | Simple approximate lookup |
| `INDEX(array, row, [col])` | Returns value at position |
| `MATCH(value, array, [match_type])` | Returns position of value |
| `INDEX/MATCH` combo | Flexible two-way lookup |
| `OFFSET(reference, rows, cols, [h], [w])` | Returns a shifted range |
| `INDIRECT(ref_text)` | Converts text to a reference |
| `CHOOSE(index, val1, val2, ...)` | Picks value by position |
| `ROW()` / `COLUMN()` | Returns row/column number |
| `ROWS(range)` / `COLUMNS(range)` | Counts rows/columns |
| `ADDRESS(row, col)` | Builds a cell reference as text |
| `FILTER(array, include, [if_empty])` | Filters data dynamically (365) |
| `SORT(array, [sort_index], [order])` | Sorts data dynamically (365) |
| `SORTBY(array, by_array, [order])` | Sorts by another array (365) |

| Formula | Purpose |
| --- | --- |
| `UNIQUE(array)` | Returns unique values (365) |

## 6. Date & Time

| Formula | Purpose |
| --- | --- |
| `TODAY()` / `NOW()` | Current date / date-time |
| `DATE(year, month, day)` | Builds a date |
| `DATEVALUE(text)` | Converts text to date |
| `DAY(date)` / `MONTH(date)` / `YEAR(date)` | Extract date parts |
| `WEEKDAY(date, [type])` | Day of week number |
| `WEEKNUM(date)` | Week number of year |
| `EDATE(start_date, months)` | Date N months away |
| `EOMONTH(start_date, months)` | Last day of month N months away |
| `DATEDIF(start, end, unit)` | Difference between dates |
| `NETWORKDAYS(start, end, [holidays])` | Working days between dates |
| `WORKDAY(start, days, [holidays])` | Date N working days away |
| `TIME(hour, min, sec)` | Builds a time value |
| `HOUR(time)` / `MINUTE(time)` / `SECOND(time)` | Extract time parts |

## 7. Financial

| Formula | Purpose |
|---|---|
| `PMT(rate, nper, pv, [fv], [type])` | Loan/investment payment |
| `FV(rate, nper, pmt, [pv], [type])` | Future value |
| `PV(rate, nper, pmt, [fv], [type])` | Present value |
| `NPV(rate, value1, value2, ...)` | Net present value |
| `IRR(values, [guess])` | Internal rate of return |
| `RATE(nper, pmt, pv, [fv], [type])` | Interest rate per period |
| `NPER(rate, pmt, pv, [fv], [type])` | Number of periods |
| `SLN(cost, salvage, life)` | Straight-line depreciation |
| `DB(cost, salvage, life, period)` | Declining balance depreciation |

## 8. Information

| Formula | Purpose |
|---|---|
| `ISNUMBER(value)` / `ISTEXT(value)` | Type checks |
| `ISBLANK(value)` | Checks empty cell |
| `ISERROR(value)` / `ISNA(value)` | Error checks |
| `ISFORMULA(reference)` | Checks if cell has formula |
| `TYPE(value)` | Returns data type code |
| `CELL(info_type, reference)` | Returns cell info |
| `N(value)` | Converts to number |
| `NA()` | Returns #N/A error |

## 9. Array / Dynamic Array Functions (Excel 365)

| Formula | Purpose |
|---|---|
| `SEQUENCE(rows, [cols], [start], [step])` | Generates a sequence of numbers |
| `RANDARRAY(rows, [cols], [min], [max])` | Array of random numbers |
| `LET(name1, value1, ..., calculation)` | Names values inside a formula |
| `LAMBDA(param, calculation)` | Defines a custom reusable function |
| `MAP(array, lambda)` | Applies a function to each array item |
| `REDUCE(initial, array, lambda)` | Reduces array to a single value |
| `BYROW(array, lambda)` / `BYCOL(array, lambda)` | Applies function by row/column |

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## 10. Data Validation "Rules" (Data tab → Data Validation)

| Rule type | What it does |
|---|---|
| Whole number / Decimal | Restricts entry to numbers in a range |
| List | Restricts entry to a dropdown list (from a range or typed list) |
| Date / Time | Restricts entry to a date/time range |
| Text length | Restricts number of characters |
| Custom formula | Any formula returning TRUE/FALSE, e.g. `=AND(A1>0,A1<100)` |
| Input message | Shows a tooltip when the cell is selected |
| Error alert | Stop / Warning / Information message on invalid entry |

## 11. Conditional Formatting "Rules" (Home → Conditional Formatting)

| Rule type | What it does |
| --- | --- |
| Highlight Cell Rules | Greater than / Less than / Between / Equal to / Text that contains / Date occurring / Duplicate values |
| Top/Bottom Rules | Top 10 items/%, Bottom 10 items/%, Above/Below average |
| Data Bars | In-cell bar proportional to value |
| Color Scales | 2- or 3-color gradient based on value |
| Icon Sets | Arrows/traffic lights/ratings based on value thresholds |
| New Rule → Formula | Custom formula-driven formatting, e.g. `=$B2<0` to highlight negatives |
| Stop If True | Halts evaluation of lower-priority rules once matched |
| Manage Rules | Reorder, edit, delete, and control rule scope/precedence |

## 12. Other Common "Rules" in Excel

| Feature | What it controls |
| --- | --- |
| Sort & Filter rules | Custom sort order, multi-level sorts, filter by color/value |
| Freeze Panes | Keeps rows/columns visible while scrolling |
| Named Ranges | Assigns a name to a cell/range for use in formulas |
| Table structured references | `Table1[ColumnName]` syntax inside Excel Tables |
| Circular reference warning | Alerts when a formula refers to its own cell |
| Iterative calculation setting | Allows controlled circular references (File → Options → Formulas) |
| Protect Sheet/Workbook rules | Locks cells, restricts editing by rule |
| AutoFill/Flash Fill patterns | Auto-detects and continues data patterns |

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### Quick Formula-Writing Rules

1. Every formula starts with `=`.
2. Use `$` to lock references: `$A$1` (absolute), `A$1`/`$A1` (mixed), `A1` (relative).

3. Wrap multi-condition logic in `IFS`/`AND`/`OR` rather than deeply nested `IF`.
4. Prefer `XLOOKUP`/`INDEX+MATCH` over `VLOOKUP` for flexibility (no fixed column order, works right-to-left).
5. Wrap volatile/error-prone formulas in `IFERROR(...)`.
6. Use structured Table references (`Table1[Sales]`) instead of hardcoded ranges when working with growing datasets.
7. Array/dynamic-array formulas (`FILTER`, `SORT`, `UNIQUE`, `SEQUENCE`) auto-spill into adjacent cells in Excel 365 — no `Ctrl+Shift+Enter` needed.